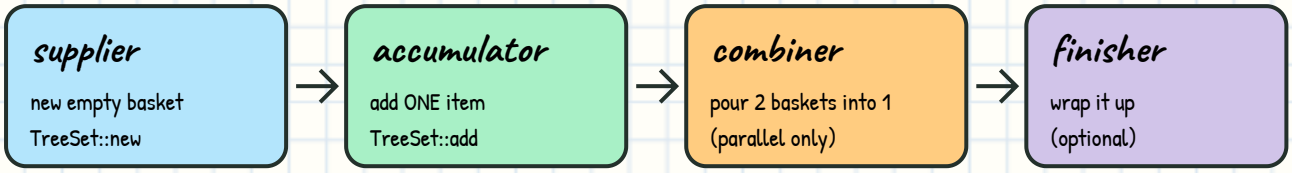
Collectors · my notes L10

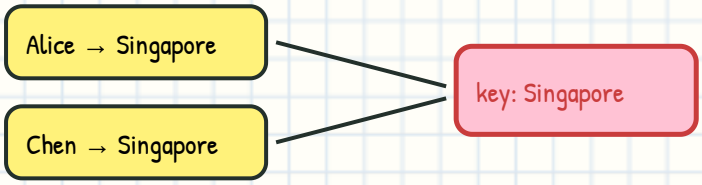
a collector is a recipe for filling a container

The 4-step recipe 🧺



→ [Bangkok, Colombo, Jakarta, Kuala Lumpur, Singapore]

toMap: two keys collide 💥



no merge → IllegalStateException

(a, b) → a + ", " + b

→ Alice Tan, Chen Wei, Ethan Lim, Jasmine Ong

null VALUE? → NullPointerException too (Map.merge)

Pick the map 🗺

TreeMap::new → sorted
LinkedHashMap::new → first seen
EnumMap → LIFE, HEALTH, MOTOR ...
default HashMap → no order

Money 💰

`reducing(ZERO, amount, add)`
→ 4429.84 for 12 payments
summingDouble ×

Collections 📦

`toSet()` → 5 cities, order? not defined
`toCollection(TreeSet::new)` → sorted
`toUnmodifiableSet()` [10] → add() 💥

Top level vs downstream

`collect(counting())` = `count()`
so why? because collectors **nest**:
groupingBy(city, **counting()**) → L11

Aggregate and compose

`joining(", ", "Flights: [", "]")`

`summarizingInt(seats)` 8 · 2078 · 160 · 471 · 259.75

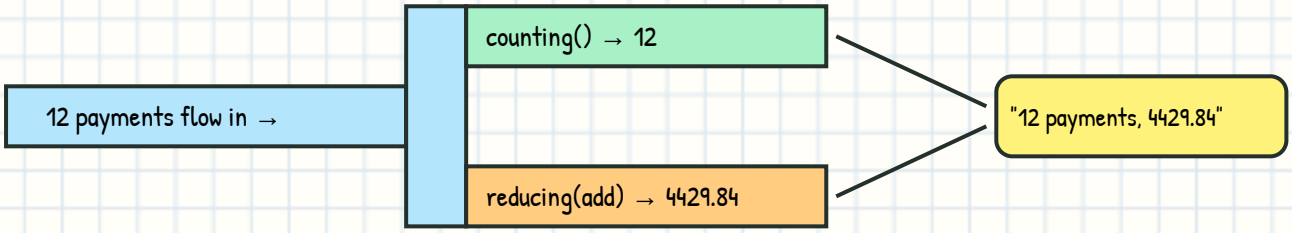
`maxBy(amount)` → P005 1500.00

`collectingAndThen` → unwrap the Optional

`filtering(SUCCESS, counting())` [9] → 8

`flatMapping` [9] → 8 Singapore accounts

teeing [12] = a T-pipe 🚰



Worked example: portfolio summary

3 hand-synced HashMaps → one `toMap(type, TypeSummary::of, TypeSummary::merge, EnumMap)` + teeing for the total · LIFE 2 / 4250.00 · … · TOTAL 10 / 10826.50 · same result in parallel ✓

Exercise: gateway per method 💳

{CARD=5, PAYNOW=4, PAYPAL=3} · {CARD=2070.29, PAYNOW=1824.30, PAYPAL=535.25} · "12 payments, 4429.84 total, failed: P003, P012"

★ Remember

recipe = supplier, accumulator, combiner, finisher · toMap: merge unless unique · choose the map · money = reducing · teeing = 2 answers, 1 pass · next → L11 groupingBy (the most important!)